

UNIT IV ASSIGNMENT

Class #2:

Rectangular Matrices with orthonormal columns

1) Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

(a) Find an orthonormal basis of the null space of A

(c) Find an orthonormal basis of the row space of A

2) For the space R^4 , let $\omega_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$, $\omega_2 = \begin{pmatrix} 3 \\ 3 \\ -1 \\ -1 \end{pmatrix}$ and $y = \begin{pmatrix} 6 \\ 0 \\ 2 \\ 0 \end{pmatrix}$ and let $\omega = \text{span}\{\omega_1, \omega_2\}$

a) Find the basis for w consists of two orthogonal vectors

b) Express y as the sum of the vector in w and a vector in $\omega^\perp$

Answer: Orthogonal basis for W is $\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ -2 \\ -2 \end{pmatrix}$, $w = \begin{pmatrix} 3 \\ 3 \\ 1 \\ 1 \end{pmatrix}$, $\omega^\perp = \begin{pmatrix} 3 \\ -3 \\ 1 \\ -1 \end{pmatrix}$

3) Obtain an orthogonal basis for the subspace of R^4 spanned by $x_1 = (1, 0, 1, 0)$, $x_2 = (1, 1, 1, 1)$, $x_3 = (-1, 2, 0, 1)$.

Answer: $(1, 0, 1, 0)$, $(0, 1, 0, 1)$, $\left(\frac{-1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{-1}{2}\right)$

4) Suppose (x_1, x_2, x_3) where $x_1 = (1, 2, 3, 0)$, $x_2 = (1, 2, 0, 0)$ and $x_3 = (1, 0, 0, 1)$ is a basis for a subspace W in R^4 . Find the orthogonal basis for w.

Answer: v_1, v_2, v_3 $(1 \ 2 \ 3 \ 0)$, $(9 \ 18 \ -15 \ 0)$, $(4 \ -2 \ 0 \ 5)$